

Untitled

```
#!pip install pandas
```

```
# import libraries for loading dataset
import pandas as pd
import numpy as np
# import libraries for plotting
import matplotlib.pyplot as plt
import seaborn as sns
from matplotlib import rcParams
%matplotlib inline
plt.style.use('dark_background')
# ignore warnings
import warnings
warnings.filterwarnings('ignore')
```

```
# loading the data set
df = pd.read_csv("adult.csv", encoding='latin-1')
print(df.head())
```

	age	workclass	fnlwgt	education	education.num	marital.status	\
0	90	?	77053	HS-grad	9	Widowed	
1	82	Private	132870	HS-grad	9	Widowed	
2	66	?	186061	Some-college	10	Widowed	
3	54	Private	140359	7th-8th	4	Divorced	
4	41	Private	264663	Some-college	10	Separated	

	occupation	relationship	race	sex	capital.gain	\
0	?	Not-in-family	White	Female	0	
1	Exec-managerial	Not-in-family	White	Female	0	
2	?	Unmarried	Black	Female	0	
3	Machine-op-inspct	Unmarried	White	Female	0	

4	Prof-specialty	Own-child	White	Female	0
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	capital.loss	hours.per.week	native.country	income
0	4356	40	United-States	<=50K
1	4356	18	United-States	<=50K
2	4356	40	United-States	<=50K
3	3900	40	United-States	<=50K
4	3900	40	United-States	<=50K

```
df.shape
```

```
(32561, 15)
```

```
df.columns
```

```
Index(['age', 'workclass', 'fnlwgt', 'education', 'education.num',
       'marital.status', 'occupation', 'relationship', 'race', 'sex',
       'capital.gain', 'capital.loss', 'hours.per.week', 'native.country',
       'income'],
      dtype='object')
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 32561 entries, 0 to 32560
Data columns (total 15 columns):
#   Column              Non-Null Count  Dtype
---  -
0   age                 32561 non-null  int64
1   workclass           32561 non-null  object
2   fnlwgt              32561 non-null  int64
3   education           32561 non-null  object
4   education.num       32561 non-null  int64
5   marital.status      32561 non-null  object
6   occupation          32561 non-null  object
7   relationship        32561 non-null  object
8   race                32561 non-null  object
9   sex                 32561 non-null  object
10  capital.gain        32561 non-null  int64
11  capital.loss        32561 non-null  int64
```

```

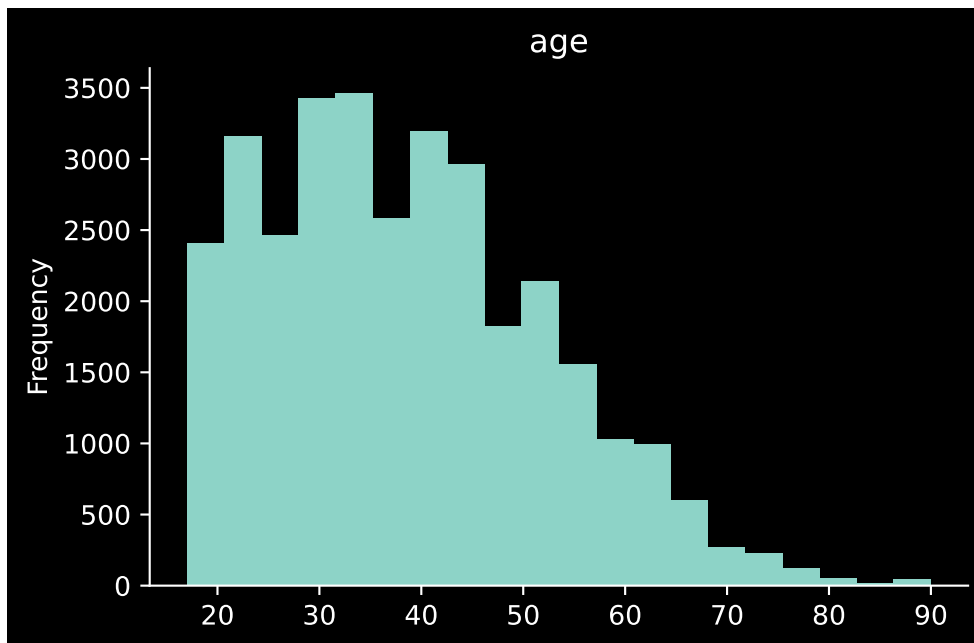
12  hours.per.week  32561 non-null  int64
13  native.country  32561 non-null  object
14  income          32561 non-null  object
dtypes: int64(6), object(9)
memory usage: 3.7+ MB

```

```

from matplotlib import pyplot as plt
df['age'].plot(kind='hist', bins=20, title='age')
plt.gca().spines[['top', 'right']].set_visible(False)
plt.show()

```



```
print(df.tail())
```

	age	workclass	fnlwgt	education	education.num	marital.status	\
32556	22	Private	310152	Some-college	10	Never-married	
32557	27	Private	257302	Assoc-acdm	12	Married-civ-spouse	
32558	40	Private	154374	HS-grad	9	Married-civ-spouse	
32559	58	Private	151910	HS-grad	9	Widowed	
32560	22	Private	201490	HS-grad	9	Never-married	

	occupation	relationship	race	sex	capital.gain	\
32556	Protective-serv	Not-in-family	White	Male	0	
32557	Tech-support	Wife	White	Female	0	

32558	Machine-op-inspct	Husband	White	Male	0
32559	Adm-clerical	Unmarried	White	Female	0
32560	Adm-clerical	Own-child	White	Male	0

	capital.loss	hours.per.week	native.country	income
32556	0	40	United-States	<=50K
32557	0	38	United-States	<=50K
32558	0	40	United-States	>50K
32559	0	40	United-States	<=50K
32560	0	20	United-States	<=50K

```
# As we observe education and education_num both are same , so we can drop one of the column
df.drop(['education.num'], axis = 1, inplace = True)
df.drop(['race'], axis = 1, inplace = True)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 32561 entries, 0 to 32560
Data columns (total 13 columns):
#   Column                Non-Null Count  Dtype
---  -
0   age                    32561 non-null  int64
1   workclass              32561 non-null  object
2   fnlwgt                 32561 non-null  int64
3   education              32561 non-null  object
4   marital.status         32561 non-null  object
5   occupation             32561 non-null  object
6   relationship           32561 non-null  object
7   sex                    32561 non-null  object
8   capital.gain           32561 non-null  int64
9   capital.loss           32561 non-null  int64
10  hours.per.week         32561 non-null  int64
11  native.country         32561 non-null  object
12  income                 32561 non-null  object
dtypes: int64(5), object(8)
memory usage: 3.2+ MB
```

```
print(df.describe().T)
```

count	mean	std	min	25%	\
-------	------	-----	-----	-----	---

age	32561.0	38.581647	13.640433	17.0	28.0
fnlwgt	32561.0	189778.366512	105549.977697	12285.0	117827.0
capital.gain	32561.0	1077.648844	7385.292085	0.0	0.0
capital.loss	32561.0	87.303830	402.960219	0.0	0.0
hours.per.week	32561.0	40.437456	12.347429	1.0	40.0

	50%	75%	max
age	37.0	48.0	90.0
fnlwgt	178356.0	237051.0	1484705.0
capital.gain	0.0	0.0	99999.0
capital.loss	0.0	0.0	4356.0
hours.per.week	40.0	45.0	99.0

```
df.nunique()
```

```
age          73
workclass     9
fnlwgt    21648
education    16
marital.status  7
occupation   15
relationship   6
sex           2
capital.gain  119
capital.loss   92
hours.per.week  94
native.country  42
income        2
dtype: int64
```

```
df['age'].value_counts()
```

```
age
36    898
31    888
34    886
23    877
35    876
...
83     6
88     3
```

```
85      3
86      1
87      1
Name: count, Length: 73, dtype: int64
```

```
df['workclass'].value_counts()
colors = ["black","red", "green", "blue", "orange", "yellow", "purple"]
df.workclass.value_counts().plot.bar(color=colors,legend=True)
plt.show()
```

